A Survey on Machine Learning and Deep Learning Approaches for DDoS Attack Detection: Methods, Datasets, and Emerging Applications

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***Abstract*—Abstract— Distributed Denial of Service (DDoS) attacks remain one of the most critical threats to modern networked systems, causing severe service disruptions, financial losses, and reputational damage across cloud computing environments. This survey provides a comprehensive examination of machine learning (ML) and deep learning (DL) techniques applied to DDoS attack detection across diverse network settings. It explores recent advancements in supervised, unsupervised, and other advanced learning approaches. Additionally, the survey reviews commonly used benchmark datasets and evaluates feature engineering and feature selection strategies. It also discusses the practical challenges encountered when deploying these detection systems in real-world environments.**

***Index Terms*—DDoS attacks, machine learning, deep learning, intrusion detection, ensemble learning, network security**

I. Introduction

Distributed Denial of Service (DDoS) attacks have evolved into one of the most pervasive and damaging cyber threats facing modern network infrastructures. These attacks aim to overwhelm target systems—servers, networks, or applications—with an excessive volume of malicious traffic, rendering them unavailable to legitimate users and causing substantial financial and operational damage [1], [2]. Unlike simple Denial of Service (DoS) attacks that originate from a single source, DDoS attacks leverage distributed botnets comprising thousands of compromised devices to generate coordinated flooding traffic, making detection and mitigation significantly more challenging.

The threat landscape has dramatically intensified in recent years. Reports indicate that DDoS incidents continue to escalate, with attack volumes growing by more than 50% in 2024 and peak capacities exceeding 1.6 Tbps [4]. Furthermore, a 358% year-over-year increase in blocked DDoS attacks has been documented, with hyper-volumetric attacks exceeding 6.5 terabits per second [5]. The emergence of massive botnets—one comprising over 13,000 IoT devices for a single record-breaking attack—demonstrates the capacity of threat actors to overwhelm even well-provisioned defenses [5].

Traditional approaches to DDoS detection rely on signature-based or rule-based Intrusion Detection Systems (IDS) that compare incoming network traffic against predefined patterns. While effective against known attack signatures, these conventional methods are fundamentally limited in their ability to detect novel, zero-day, or obfuscated attacks that evolve to evade static detection rules [4]. The sheer volume and complexity of modern network traffic, amplified by cloud services, IoT proliferation, and 5G networks, further strain traditional detection mechanisms that cannot scale to process high-throughput data streams in real time.

This limitation has driven a paradigm shift toward data-driven detection methodologies, particularly Machine Learning (ML) and Deep Learning (DL) techniques. ML models can learn complex patterns from network traffic data, classify incoming flows as benign or malicious, and adapt to evolving threat landscapes without requiring manually crafted signatures [1]. Deep learning architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory (LSTM) networks, offer superior alternatives by automatically extracting hierarchical feature representations from raw traffic data [5].

The convergence of DDoS threats with emerging network paradigms has created new attack surfaces and defense challenges.Cloud computing environments face unique challenges where elastic resource scaling and inter-VM traffic amplification complicate detection [3]. Autonomous vehicles, as safety-critical cyber-physical systems, face existential threats from DDoS attacks that could disrupt real-time navigation and communication [5].

A. Motivation and Scope

Despite extensive research on DDoS detection, several critical gaps persist in the literature. First, many existing ML models are trained and evaluated exclusively on binary classification tasks (benign vs. malicious), leaving the more challenging multiclass problem—distinguishing among heterogeneous DDoS attack vectors—only partially addressed [4]. Second, most studies focus on achieving high accuracy on static benchmark datasets while overlooking adversarial robustness and temporal variability, which are essential for real-world deployment [3]. Third, the rapid diversification of network environments (cloud, autonomous vehicles) demands domain-specific detection approaches that account for unique traffic characteristics and operational constraints .

This survey aims to address these gaps by providing a comprehensive, up-to-date examination of ML and DL approaches for DDoS detection across diverse network environments. We synthesize findings from recent studies published between 2024 and 2026, critically analyze the strengths and limitations of various detection methodologies, and identify open challenges and future research directions.

B. Contributions

The major contributions of this survey are:

1) A comprehensive review of DDoS attack types, their characteristics, and the evolving threat landscape across cloud,and autonomous vehicle environments.
2) A systematic analysis of benchmark datasets used for DDoS detection research, including their strengths, limitations, and suitability for different evaluation scenarios.
3) A detailed examination and comparison of ML and DL techniques—including ensemble methods, hybrid models, and federated learning—applied to DDoS detection, with emphasis on their detection performance, computational efficiency, and real-world applicability.
4) Identification of open challenges—adversarial robustness, multiclass detection, real-time constraints, and cross-domain generalization—and future research directions to guide the development of next-generation DDoS detection systems.

C. Paper Organization

The remainder of this paper is organized as follows. Section II provides background on DDoS attacks, their taxonomy, and the evolution of detection approaches. Section III reviews the benchmark datasets commonly used in DDoS detection research. Section IV examines ML-based detection methods, including traditional ML algorithms and ensemble approaches. Section V covers deep learning approaches and advanced architectures.Section VI analyzes comparative performance across reviewed studies. Section VII identifies open challenges and future research directions. Section VIII concludes the paper.

II. Background: DDoS Attacks and Detection Approaches

A. Taxonomy of DDoS Attacks

DDoS attacks can be broadly categorized based on the network layer they target and the resources they aim to exhaust. Understanding this taxonomy is essential for designing effective detection mechanisms.

Volumetric Attacks represent the most common category, aiming to saturate the bandwidth of the target network by flooding it with massive volumes of traffic. Common examples include UDP floods, ICMP floods, and DNS amplification attacks. These attacks leverage botnets to generate traffic volumes that can exceed terabits per second, overwhelming network infrastructure and upstream links [5].

Protocol Attacks (also called state-exhaustion attacks) exploit weaknesses in network protocol implementations to consume server resources or intermediate communication equipment such as firewalls and load balancers. SYN flood attacks, which exploit the TCP three-way handshake mechanism, are prototypical examples. These attacks exhaust connection state tables, preventing legitimate connections from being established [1].

Application-Layer Attacks target specific application services (typically HTTP/HTTPS) with seemingly legitimate requests designed to exhaust server processing capacity. These attacks are particularly challenging to detect because the traffic closely resembles normal user behavior, operating at lower volumes but consuming disproportionate computational resources [1].

Low-Rate DDoS Attacks represent a sophisticated variant that uses relatively small amounts of malicious traffic to exploit specific vulnerabilities or protocol limitations within the target infrastructure. Unlike high-volume attacks, low-rate attacks are designed to degrade service quality while remaining below traditional detection thresholds, making them particularly difficult to identify [1].

In cloud computing environments, DDoS attacks present additional complexity due to the multi-tenant, virtualized nature of the infrastructure. Attacks targeting cloud Virtual Machines (VMs) can exploit resource contention among co-located VMs, where overwhelming one VM's resources affects others sharing the same physical host [3]. Advanced adversaries may also target the hypervisor layer, threatening VM isolation and amplifying the impact across the cloud infrastructure.

B. Evolution of Detection Approaches

The evolution of DDoS detection can be characterized in three phases. **First-generation** approaches relied on static, signature-based detection using predefined rules to match known attack patterns. While effective for known threats, these methods are inherently reactive and cannot detect novel attack variants.

Second-generation approaches introduced statistical and anomaly-based detection methods, including threshold-based monitoring, entropy analysis, and Change Point Detection (CPD) algorithms such as CUSUM and EWMA [1]. These methods can identify deviations from normal traffic behavior but often suffer from high false positive rates when network conditions fluctuate legitimately.

Third-generation approaches leverage ML and DL techniques that learn traffic patterns from data, enabling adaptive detection of both known and unknown attacks. These approaches have demonstrated superior performance across multiple metrics and form the primary focus of this survey [1].

III. Benchmark Datasets for DDoS Detection

The quality and representativeness of training data fundamentally determines the effectiveness of ML-based DDoS detection systems. This section reviews the most widely used benchmark datasets in the literature.

A. CIC-DDoS2019

The CIC-DDoS2019 dataset, developed by the Canadian Institute for Cybersecurity, is among the most widely adopted benchmarks for DDoS detection research. It contains approximately 8.85 million flow records encompassing both benign traffic and multiple DDoS attack types including DNS, LDAP, MSSQL, NetBIOS, NTP, SNMP, SSDP, SYN, TFTP, UDP, and UDP-Lag attacks [4], [5]. The dataset provides over 80 extracted features per flow, covering packet-level statistics, flow duration, byte counts, and inter-arrival times.

Multiple studies in our review utilize this dataset. Samy et al. [3] employed CIC-DDoS2019 to evaluate their Optimized CatBoost model, achieving 99.2% accuracy. Angulo et al. [4] used it for multiclass detection with nine attack categories, achieving 91.26% accuracy. Djenna et al. [5] achieved 99.98% accuracy using deep learning with federated learning on this dataset.

While CIC-DDoS2019 offers diverse attack types and large-scale traffic, it has notable limitations: the data was generated in a controlled testbed environment that may not fully capture real-world network dynamics, and class imbalance between attack categories can bias model training [4].

B. CIC-IDS2017 and NSL-KDD

The CIC-IDS2017 dataset provides network traffic captures including brute force, DoS, DDoS, web attacks, and infiltration scenarios. It has been widely used in intrusion detection research but contains a more limited range of DDoS-specific attack types compared to CIC-DDoS2019 [**?**], [1].

The NSL-KDD dataset, an improved version of the original KDD Cup 1999 dataset, remains referenced in some studies despite being considered outdated for modern DDoS detection. It lacks SDN-specific features and does not reflect contemporary attack patterns, reducing its applicability to current network environments.

C. Dataset Challenges

A critical challenge across all datasets is the gap between laboratory-generated traffic and real-world production network conditions. As noted by Tebbaa et al. [**?**], many studies rely on inappropriate datasets that do not reflect the complexity and diversity of modern network environments. Key dataset challenges include:

- **Class imbalance**: Attack traffic often vastly outnumbers benign traffic (or vice versa), biasing model training and evaluation.
- **Temporal staleness**: Older datasets do not capture modern attack patterns, encrypted traffic, or contemporary network architectures.
- **Scale limitations**: Many datasets are generated in small-scale testbeds that do not replicate production-level traffic volumes and patterns.

IV. Machine Learning Approaches for DDoS Detection

This section examines traditional machine learning algorithms and ensemble methods applied to DDoS detection. Table I provides a comparative summary of the ML and DL approaches reviewed in this survey.

A. Traditional ML Algorithms

K-Nearest Neighbor (KNN): KNN is a non-parametric, instance-based learning algorithm that classifies samples based on the majority class among their k nearest neighbors in the feature space. Raja et al. demonstrated that KNN achieved 97.13% accuracy for DDoS detection in smart home environments using real traffic data, outperforming the ANN model (81.7%) under the same conditions. The simplicity and interpretability of KNN make it suitable for resource-constrained IoT environments. Raghavendran and Robinson reported 97% accuracy with KNN in their WISNE-SDN framework for IoT-based DDoS detection.

Naive Bayes (NB): Naive Bayes classifiers apply Bayes' theorem with strong independence assumptions between features. While computationally efficient with minimal training requirements, NB typically demonstrates lower accuracy compared to tree-based methods for DDoS detection tasks due to the conditional independence assumption being violated in network traffic features.

Support Vector Machines (SVM): SVMs construct optimal hyperplanes to separate classes in high-dimensional feature spaces. While SVMs can achieve strong performance in binary classification scenarios, they face scalability challenges with large datasets and struggle in dynamic, high-dimensional traffic environments where the decision boundary is complex [4].

Decision Trees (DT): Decision tree algorithms recursively partition the feature space based on information gain or Gini impurity criteria. They offer high interpretability and fast inference but are prone to overfitting on training data. DTs serve as foundational building blocks for more powerful ensemble methods [1].

B. Ensemble Learning Methods

Ensemble methods have emerged as the dominant approach for DDoS detection, consistently achieving the highest accuracy across multiple studies. These methods combine multiple base learners to produce predictions that are more robust and accurate than any individual model.

Random Forest (RF): Random Forest constructs multiple decision trees trained on bootstrapped subsets of the data and aggregates their predictions through majority voting. RF is widely adopted for DDoS detection due to its robustness to overfitting, ability to handle high-dimensional data, and inherent feature importance estimation.

XGBoost (XGB): Extreme Gradient Boosting is a regularized gradient boosting framework that builds an additive ensemble of decision trees, with each tree correcting the errors of its predecessors. XGBoost has demonstrated exceptional performance in DDoS detection.Its computational efficiency and built-in regularization make it well-suited for high-throughput detection scenarios.

CatBoost: CatBoost is a gradient boosting algorithm that natively handles categorical features and employs ordered boosting to reduce overfitting. Samy et al. [3] proposed an Optimized CatBoost ML (OCML) framework with Optuna-based Bayesian hyperparameter optimization and SHAP-driven feature selection, achieving 99.2% accuracy on CIC-DDoS2019. Notably, their study also evaluated adversarial robustness, reporting accuracies of 97%, 80%, and 71% against FGSM, Carlini-Wagner, and PGD attacks respectively.

C. Hybrid and Stacking Ensemble Models

Several studies have explored hybrid architectures that combine complementary classifiers to improve detection performance.

The model integrates Random Forest for its robustness in handling diverse feature sets with XGBoost's gradient boosting capability for error correction. Evaluated on a custom SDN dataset with features extracted from OpenFlow monitoring messages, the hybrid model achieved 99.36% accuracy with near-perfect ROC AUC discrimination, demonstrating clear superiority over individual classifiers.

Angulo et al. [4] introduced a more sophisticated three-layer hierarchical stacking ensemble for multiclass DDoS detection. Their architecture comprises:

- **Layer Zero (Base Learners):** Heterogeneous S-ML and D-ML models including tree-based, kernel-based, and neural architectures.
- **Layer One (Meta Learners):** Regression-based and neural models trained on meta-features derived from base-layer predictions.
- **Layer Two (Final Voting):** A voting mechanism aggregating meta-model outputs for final classification.

This framework achieved 91.26% accuracy across nine attack categories on CIC-DDoS2019, representing a more realistic evaluation than binary or reduced-class settings. The lower accuracy compared to binary classification studies highlights the inherent difficulty of multiclass DDoS detection, where overlapping traffic patterns and label imbalance between attack categories present significant challenges.

D. Feature Engineering and Selection

Feature engineering and selection are critical preprocessing steps that significantly impact detection performance. The CIC-DDoS2019 dataset contains over 80 features, making dimensionality reduction essential for computational efficiency and model generalization.

Samy et al. [3] employed SHAP (SHapley Additive exPlanations) for interpretable feature selection, identifying the most influential features for DDoS detection while providing model transparency. This approach reduces dimensionality while maintaining detection accuracy and offers insights into which traffic characteristics are most indicative of attack behavior.

Mahar et al. developed SDN-specific features from OpenFlow monitoring messages, including flow and port-level statistics that capture the unique behavior of SDN-based traffic. This domain-specific feature engineering approach addresses the limitation that general-purpose features may not adequately represent SDN traffic characteristics.

V. Deep Learning Approaches for DDoS Detection

Deep learning methods have gained significant attention for DDoS detection due to their ability to automatically learn hierarchical feature representations from raw or minimally processed network traffic data, eliminating the need for extensive manual feature engineering.

A. Convolutional Neural Networks (CNNs)

CNNs apply convolutional filters to extract spatial patterns from input data. In the context of DDoS detection, CNNs can identify local patterns and correlations within network traffic features. Djenna et al. [5] implemented CNNs as one of their candidate architectures for DDoS detection in autonomous vehicle networks. Their evaluation demonstrated that while CNNs can effectively capture spatial correlations in traffic features, they require careful architecture design to achieve optimal performance across different attack types.

B. Recurrent Neural Networks (RNNs) and LSTMs

RNNs and their variants, particularly Long Short-Term Memory (LSTM) networks, are designed to capture temporal dependencies in sequential data. Network traffic inherently exhibits temporal patterns, making RNNs particularly suitable for detecting time-series anomalies associated with DDoS attacks. Djenna et al. [5] compared RNN architectures against CNNs and DNNs, finding that architectures capturing temporal relationships demonstrated strong detection capabilities for evolving attack patterns.

C. Deep Neural Networks (DNNs)

Standard feedforward Deep Neural Networks with multiple hidden layers can learn complex non-linear mappings between traffic features and attack classifications. Djenna et al. [5] found that their DNN architecture, enhanced with an active learning component for data efficiency, achieved the highest performance among the architectures tested, with 99.98% accuracy on CIC-DDoS2019.

D. Federated Learning for Privacy-Preserving Detection

Federated learning has emerged as a promising paradigm for collaborative DDoS detection that addresses privacy and data sovereignty concerns. Instead of centralizing training data, federated learning enables multiple clients to collaboratively train a shared model while keeping data locally.

Djenna et al. [5] deployed their best-performing DNN model within a federated learning framework for DDoS detection in autonomous vehicle networks. Their approach enables collaborative, privacy-preserving training across distributed clients while maintaining detection accuracy of 99.98%. This architecture is particularly relevant for autonomous vehicles, where safety-critical detection must be performed without sharing sensitive location and navigation data across organizational boundaries.

The federated learning approach also addresses the challenge of heterogeneous data distributions across different network segments or organizations, potentially improving model generalization to diverse traffic patterns encountered in real-world deployments.

E. Active Learning for Data Efficiency

Active learning strategies aim to maximize model performance while minimizing the amount of labeled training data required. Djenna et al. [5] integrated active learning with their deep learning framework, enabling the model to intelligently select the most informative samples for labeling. This approach is particularly valuable in DDoS detection scenarios where labeled attack data may be scarce or expensive to obtain, and where the cost of mislabeling benign traffic as malicious (or vice versa) is high.

VI. Emerging Application Domains

DDoS detection research has expanded beyond traditional network security into specialized application domains, each presenting unique challenges and requirements.

A. Cloud Computing Environments

Cloud environments face multi-vector DDoS attacks that simultaneously target multiple layers of the network stack. The elastic resource scaling, virtualization, and multi-tenancy characteristics of cloud infrastructure create both vulnerabilities and opportunities for detection [3].

Samy et al. [3] specifically addressed DDoS detection in cloud virtual machines, proposing the OCML framework that integrates hyperparameter optimization, explainable feature selection, and adversarial training. Their unique contribution was evaluating robustness against both adversarial ML attacks (FGSM, CW, PGD) and time-series network traffic attacks (pulse wave, random burst, slow ramp), achieving 80%, 83%, and 77% accuracy respectively for the temporal attacks. This dual robustness evaluation addresses a critical gap, as most studies evaluate only static detection accuracy without considering how adversaries might adapt their strategies to evade ML-based detectors.

B. Autonomous Vehicle Networks

Autonomous vehicles represent a safety-critical application domain where DDoS attacks can have life-threatening consequences. These cyber-physical systems rely on continuous data exchange for navigation, obstacle detection, and vehicle-to-vehicle communication, making network availability essential for safe operation [5].

Djenna et al. [5] proposed a novel deep learning framework specifically designed for DDoS detection in automotive networks. Their approach addresses three primary attack vectors: volumetric, state-exhaustion, and amplification attacks. The integration of active learning for data efficiency and federated learning for privacy-preserving collaborative training makes this framework particularly suited to the distributed, privacy-sensitive nature of autonomous vehicle ecosystems. The 99.98% detection accuracy on CIC-DDoS2019 demonstrates the feasibility of deploying high-accuracy DDoS detection in safety-critical environments.

VII. Comparative Analysis

Table I presents a comprehensive comparison of the reviewed studies. Several important observations emerge from this analysis.

Accuracy vs. Task Complexity: Binary classification models consistently report higher accuracy (99–100%) compared to multiclass models (91.26%). Angulo et al. [4] explicitly addressed this discrepancy, noting that many prior works report near-perfect performance under binary or reduced-class settings with conditions that artificially inflate metrics. The 91.26% accuracy achieved by their multiclass stacking ensemble under realistic nine-class evaluation represents a more demanding and representative test scenario.

Ensemble Superiority: Across studies, ensemble methods (RF, XGBoost, CatBoost, and hybrid/stacking combinations) consistently outperform individual algorithms. The hybrid RF-XGBoost model and the Optimized CatBoost [3] both exceed 99% accuracy, confirming that combining complementary learning strategies produces more robust classifiers.

Deep Learning Trade-offs: Deep learning approaches achieve the highest reported accuracy (99.98% by Djenna et al. [5]) but require substantially larger datasets and computational resources. In contrast, Raja et al. demonstrated that in data-constrained smart home environments, traditional KNN (97.13%) significantly outperformed ANN (81.7%), underscoring that deep learning is not universally superior and that model selection should consider the operational context.

VIII. Open Challenges and Future Research Directions

A. Open Challenges

Adversarial Robustness: Most DDoS detection models are evaluated exclusively on static benchmark accuracy without considering adversarial attacks designed to evade detection. Samy et al. [3] demonstrated significant accuracy degradation under adversarial perturbations (from 99.2% to 71% under PGD attacks), highlighting the vulnerability of ML-based detectors to adversarial manipulation. Developing detection models that maintain high accuracy under adversarial conditions remains a critical open challenge.

Multiclass Detection: The majority of existing approaches focus on binary classification, leaving the multiclass problem insufficiently addressed. Real-world networks experience diverse attack types simultaneously, requiring detectors that can not only distinguish malicious from benign traffic but also classify specific attack categories for appropriate mitigation. As demonstrated by Angulo et al. [4], multiclass detection with nine categories remains significantly more challenging than binary classification, with accuracy dropping from the 99%+ range to approximately 91%.

TABLE I
COMPARATIVE SUMMARY OF REVIEWED DDOS DETECTION APPROACHES

Study	Method	Dataset	Domain	Accuracy	Task	Key Contribution
Samy et al. [3]	Optimized CatBoost (OCML)	CIC-DDoS2019	Cloud VMs	99.2%	Binary	Adversarial robustness & SHAP feature selection
Angulo et al. [4]	Stacking Ensemble (S-ML + D-ML)	CIC-DDoS2019	General	91.26%	Multiclass (9 classes)	Hierarchical 3-layer stacking for multiclass
Raja et al. [?]	KNN, ANN	Real smart home data	Smart Home/ IoT	97.13% (KNN)	Binary	Real-world smart home traffic evaluation
Djenna et al. [5]	CNN, RNN, DNN + Federated Learning	CIC-DDoS2019	Autonomous Vehicles	99.98%	Multiclass (3 types)	Active + federated learning for vehicles
Owusu et al. [1]	Survey (CPD + ML + Sampling)	Multiple	General	N/A	Survey	Integration of sampling, CPD, and ML

Real-Time Processing Constraints: Deploying ML-based detection in production networks requires processing high-throughput traffic streams with minimal latency. The computational demands of complex ensemble or deep learning models may not meet the real-time requirements of high-speed networks [1]. Balancing detection accuracy with inference speed remains an active area of investigation.

Dataset Representativeness: Current benchmark datasets do not adequately capture the diversity, scale, and temporal dynamics of real-world network traffic. The gap between laboratory-generated and production traffic limits the generalizability of evaluation results. Furthermore, the rapid evolution of attack strategies means that datasets become stale quickly [1].

Encrypted Traffic: The increasing prevalence of HTTPS and encrypted network traffic poses a fundamental challenge for DDoS detection systems that rely on payload inspection or application-layer feature extraction. Detection methods must increasingly rely on flow-level metadata and behavioral patterns rather than content-based features [1].

Cross-Domain Generalization: Models trained on one network environment (e.g., cloud) may not generalize effectively to other domains (e.g.,autonomous vehicles) due to differences in traffic characteristics, feature distributions, and attack vectors. Domain adaptation and transfer learning techniques remain underexplored in this context.

B. Future Research Directions

Based on our comprehensive review, we identify the following promising directions for future research:

Adversarial-Aware Detection: Integrating adversarial training and robust optimization techniques into DDoS detection pipelines, following the approach pioneered by Samy et al. [3], to develop detectors that maintain effectiveness under adversarial conditions.

Explainable AI for DDoS Detection: Incorporating interpretability techniques such as SHAP [3] and attention mechanisms to provide transparency into detection decisions, enabling network administrators to understand and trust automated detection outputs.

Federated and Collaborative Detection: Extending federated learning frameworks [5] to enable organizations to collaboratively improve detection models without sharing sensitive network data, potentially incorporating differential privacy guarantees.

Lightweight Models for Edge Deployment: Developing computationally efficient detection models suitable for deployment on resource-constrained devices such as IoT gateways, smart home controllers, and edge computing nodes, building on the findings of Raja et al. that simpler models can outperform complex ones in constrained environments.

Temporal and Streaming Detection: Advancing online detection methods that can process continuous traffic streams, adapt to concept drift, and detect attacks as they emerge rather than in batch processing mode. Integration of Change Point Detection with ML methods, as proposed by Owusu et al. [1], represents a promising direction.

Realistic and Evolving Datasets: Creating and maintaining open-source, continuously updated datasets that capture real-world traffic patterns, modern attack vectors, encrypted traffic, and diverse network architectures to better support the development and evaluation of practical detection systems.

IX. LESSONS LEARNED

Through this comprehensive survey, several key insights emerge that are instructive for both researchers and practitioners:

No single model dominates. The optimal detection approach depends heavily on the operational context. Ensemble methods excel in general-purpose scenarios with abundant data, while lightweight traditional ML models may be more appropriate for resource-constrained IoT environments. This underscores the need to match model complexity with deployment constraints.

Evaluation methodology matters. The stark accuracy difference between binary (99%+) and multiclass (91%) evaluation settings demonstrates that performance claims must be contextualized by evaluation methodology. Studies reporting near-perfect accuracy on binary tasks should be interpreted cautiously, as real-world deployments face more complex classification scenarios.

Feature engineering remains critical. Despite the promise of automated feature learning in deep learning, domain-specific feature engineering (SDN-specific, IoT-specific) consistently contributes to strong performance. Understanding the target network's characteristics and crafting relevant features remains a valuable complement to purely data-driven approaches.

Robustness is under-explored. The vast majority of studies evaluate only standard accuracy metrics on static datasets. The vulnerability exposed by adversarial attacks [3] and the challenge of temporal variability [3] indicate that current detection systems may be significantly less effective in adversarial real-world conditions than benchmark results suggest.

Real data changes the story. Studies using real-world data produce different conclusions than those using benchmark datasets, highlighting the importance of validating detection approaches on genuine production traffic.

X. Conclusion

This survey has provided a comprehensive examination of machine learning and deep learning approaches for DDoS attack detection across diverse network environments. We have systematically reviewed recent advancements spanning cloud computing, Software-Defined Networking, Internet of Things, smart home infrastructures, and autonomous vehicle networks, synthesizing findings from studies published between 2024 and 2026.

Our analysis reveals several key findings. First, ensemble learning methods—particularly hybrid combinations of Random Forest, XGBoost, and CatBoost—consistently achieve the highest detection accuracy across different domains, with reported accuracies exceeding 99% in binary classification settings. Second, the more challenging and realistic multiclass detection problem remains significantly underexplored, with the best reported multiclass accuracy at 91.26% across nine DDoS categories. Third, the choice of detection model should be informed by operational constraints: deep learning excels with large datasets and computational resources, while traditional ML models may be more appropriate for resource-constrained environments such as smart homes and IoT devices.

We have identified several critical open challenges including adversarial robustness, where ML-based detectors show significant accuracy degradation under adversarial perturbations; real-time processing constraints that limit the deployment of complex models in high-speed networks; and dataset representativeness, where current benchmarks do not adequately reflect modern traffic patterns and attack vectors.

Promising future research directions include adversarial-aware detection frameworks, federated learning for privacy-preserving collaborative detection, lightweight models for edge deployment, and the development of realistic, continuously updated benchmark datasets. The integration of explainable AI techniques to provide transparency into detection decisions and the development of unified cross-domain frameworks through transfer learning represent particularly impactful avenues for advancing the field.

As DDoS attacks continue to grow in scale, sophistication, and diversity, the need for intelligent, adaptive, and robust detection mechanisms becomes increasingly urgent. The convergence of advanced ML/DL techniques with domain-specific knowledge and emerging paradigms such as federated learning offers promising pathways toward building more resilient and effective defense systems for the evolving cyber threat landscape.

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